

# Modular forms and $\ell$ -adic representations

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# Chapter 1

## Introduction

Let

$$D(q) = q \prod_{n=1}^{\infty} (1 - q^n)^{24} = \sum_{n=1}^{\infty} \tau(n) q^n \quad (|q| < 1)$$

and

$$\Delta(z) = D(e^{2\pi iz}) \quad (\operatorname{Im}(z) > 0),$$

It is known that, up to a constant factor, the function  $\Delta$  is the unique parabolic modular form of weight 12 for the group  $SL_2(\mathbb{Z})$ .

For a prime  $p$ , define

$$H_p(X) = 1 - \tau(p)X + p^{11}X^2.$$

According to Hecke's theory, the Dirichlet series

$$L_{\tau}(s) = \sum_{n=1}^{\infty} \tau(n) n^{-s} = \prod_{p \in P} \frac{1}{H_p(p^{-s})}$$

extends to an entire function of  $s$ , and the function

$$(2\pi)^{-s} \Gamma(s) L_{\tau}(s)$$

is invariant under  $s \leftrightarrow 12 - s$ .

Ramanujan's conjecture asserts that the roots of the polynomial  $H_p$  have absolute value  $p^{-11/2}$  (i.e.,  $|\tau(p)| < 2p^{11/2}$ ).

These proven or conjectural properties are analogous to the conjectural properties of zeta functions of algebraic varieties over  $\mathbb{Q}$ . This suggests, as a first approximation, trying to interpret  $L_{\tau}$  as the zeta function of such a variety.

For each prime  $\ell$ , let  $K_{\ell}$  be the largest extension of  $\mathbb{Q}$  unramified outside  $\ell$ , and for  $p \neq \ell$ , let  $F_p$  be the inverse of the Frobenius element  $\varphi_p$  in the Galois group  $\operatorname{Gal}(K_{\ell}/\mathbb{Q})$ . The latter is well-defined up to conjugation.

Translating this into terms of  $\ell$ -adic cohomology, Serre conjectured the existence, for each  $\ell$ , of a representation of  $\operatorname{Gal}(K_{\ell}/\mathbb{Q})$  into a  $\mathbb{Q}_{\ell}$ -vector space  $V_{\ell}$  of rank 2 such that for each  $p \neq \ell$ ,

$$H_p(X) = \det(1 - F_p X; V_\ell).$$

Moreover, the representation  $V_\ell$  should fall within the scope of the Weil conjectures, making Ramanujan's conjecture a special case of the latter.

This program was successfully carried out by Kuga-Shimura [4] in the analogous case of modular forms related to certain compact quotient subgroups of  $SL_2(\mathbb{R})$ . Reduced to the present case, the fundamental idea of Sato-Kuga-Shimura is as follows: if  $E$  is the universal elliptic curve over the moduli scheme  $S$  of elliptic curves (ignoring for now that it does not exist) and if  $E^k$  is the  $k$ -fold fiber product of  $E$  with itself over  $S$ , then  $L_\tau(s)$  is essentially the zeta function of  $E^k$  for  $k = 10 = 12 - 2$ .

What follows explains how to resolve the difficulties created by the cusps and how to construct the representations  $V_\ell$  with the properties indicated above. For more historical details and applications, we refer to Serre [6].

## Notations

- Let  $\mathbb{A}$  denote the ring of adeles of  $\mathbb{Q}$ ,  $\mathbb{A}^f$  the ring of "finite" adeles, the restricted product over all primes of the fields  $\mathbb{Q}_p$ , and for  $S$  a set of primes, define

$$\mathbb{A}_S^f = \prod_{p \in S} \mathbb{Q}_p \times \prod_{p \notin S} \mathbb{Z}_p \subset \mathbb{A}^f.$$

For  $S = \emptyset$ , write  $\hat{\mathbb{Z}} = \mathbb{A}_\emptyset^f$ .

- If  $X$  is a topological space (or the étale site of a scheme) and  $G$  a set, denote by  $\underline{G}$  the constant sheaf on  $X$  defined by  $G$ .

- Let  $\mathbb{G}_a$  and  $\mathbb{G}_m$  denote the additive and multiplicative groups, respectively.

- An elliptic curve is a one-dimensional abelian variety, in particular equipped with an origin.

- If  $\mathcal{L}$  is an invertible sheaf and  $n \in \mathbb{Z}$ , denote by  $\mathcal{L}^n$  its  $n$ -th tensor power.

- Let  $\overline{\mathbb{Q}}$  denote the algebraic closure of  $\mathbb{Q}$  in  $\mathbb{C}$ .

- The symbol  $\square$  marks the end of a proof or its absence.

## Chapter 2

# The Shimura Isomorphism

### No. 2 - The Shimura Isomorphism

(2.1) An elliptic curve over a complex analytic space  $S$  is a proper and flat morphism of analytic spaces  $f : E \rightarrow S$ , equipped with a section  $e$ , whose fibers are elliptic curves. An elliptic curve over  $S$  admits a unique  $S$ -group law  $\mu : E \times_S E \rightarrow E$  with identity section  $e$ . To an elliptic curve are associated:

(a) The invertible sheaf  $\omega_E = e^* \Omega_{E/S}^1$ . The relative Lie algebra  $\underline{\text{Lie}}_S(E)$  is the invertible sheaf  $\omega^{-1}$ , dual to  $\omega$ . We have  $f_* \Omega_{E/S}^1 \cong \omega$ .

(b) The local system of free  $\mathbb{Z}$ -modules of rank 2  $R^1 f_* \mathbb{Z}$ . Set  $T_{\mathbb{Z}}(E) = R^1 f_* \mathbb{Z}^{\vee}$  and  $T_{\mathbb{Q}}(E) = T_{\mathbb{Z}}(E) \otimes \mathbb{Q}$  (local system of the homology of  $E$  over  $S$ ).

The exponential map defines an exact sequence of sheaves of sections:

$$0 \rightarrow T_{\mathbb{Z}}(E) \xrightarrow{\alpha} \omega^{-1} \rightarrow E \rightarrow 0,$$

so that the elliptic curve  $E$  is reconstructed from the map  $\alpha$ .

The local system  $\Lambda^2 R^1 f_* \mathbb{Z} \cong R^2 f_* \mathbb{Z}$  is canonically isomorphic to  $\mathbb{Z}$ . An isomorphism between  $\mathbb{Z}^2$  and  $R^1 f_* \mathbb{Z}$  is called *permitted* if it induces  $-1$  on the second exterior powers.

Let  $\text{Hom}^+(\mathbb{R}^2, \mathbb{C})$  denote the set of isomorphisms (of  $\mathbb{R}$ -vector spaces) between  $\mathbb{R}^2$  and  $\mathbb{C}$  that do *not* preserve the natural orientations of  $\mathbb{R}^2$  and  $\mathbb{C}$  (defined by  $e_1 \wedge e_2 > 0$  and  $1 \wedge i > 0$ ). Such a homomorphism is determined by its restriction to  $\mathbb{Z}^2$ , and we set

$$\text{Hom}^+(\mathbb{Z}^2, \mathbb{C}) = \text{Hom}^+(\mathbb{R}^2, \mathbb{C}).$$

This space is endowed with the complex structure induced by its inclusion into the complex vector space  $\text{Hom}(\mathbb{Z}^2, \mathbb{C})$ . Over this space, there exists a universal exact sequence:

$$0 \rightarrow \mathbb{Z}^2 \xrightarrow{\alpha} \mathbb{G}_a \rightarrow E_0 \rightarrow 0.$$

**PROPOSITION 2.2.** (i) The functor associating to each analytic space  $S$  the set of isomorphism classes of elliptic curves  $E$  over  $S$ , equipped with isomorphisms  $\omega_E \cong \mathbb{G}_a$  and  $R^1 f_* \mathbb{Z} \cong \mathbb{Z}^2$  (the latter being permitted), is represented by the analytic space  $\text{Hom}^+(\mathbb{R}^2, \mathbb{C})$ , endowed with the universal elliptic curve  $E_0$ .

(ii) The functor associating to each analytic space  $S$  the set of isomorphism classes of

elliptic curves over  $S$ , equipped with a permitted isomorphism  $R^1 f_* \underline{\mathbb{Z}} \cong \underline{\mathbb{Z}}^2$ , is represented by the analytic space  $X = \mathbb{C}^\times \backslash \text{Hom}^+(\mathbb{R}^2, \mathbb{C})$  (Poincaré upper half-plane).

(iii) The space  $\text{Hom}^+(\mathbb{R}^2, \mathbb{C})$  is a principal homogeneous space with group  $\mathbb{G}_m$  over  $X$ .  $\square$

We may also view  $X$  as the set of complex structures on  $\mathbb{R}^2$ . By (ii), it is equipped with a universal elliptic curve  $E_X$ , whose real cohomology local system is canonically isomorphic to  $\underline{\mathbb{R}}^2$ . Let  $\omega$  be the invertible sheaf associated to  $E_X$ .

The coherent analytic sheaf  $R^1 f_* \underline{\mathbb{R}} \otimes_{\mathbb{R}} \mathcal{O}_X$  is the sheaf of relative de Rham cohomology of  $E_X$  over  $X$ , fitting into an exact sequence (Hodge filtration):

$$0 \rightarrow \omega \rightarrow R^1 f_* \underline{\mathbb{R}} \otimes_{\mathbb{R}} \mathcal{O}_X \xrightarrow{q} \omega^{-1} \rightarrow 0$$

(since by Serre duality,  $\omega^{-1} \cong R^1 f_* \mathcal{O}$ ).

The functorial description 2.2(ii) makes evident a right action of the group  $SL_2(\mathbb{Z})$  on  $(X, E_X)$ : for  $\gamma \in SL_2(\mathbb{Z})$ , to the elliptic curve  $E$  with  $\alpha : \underline{\mathbb{Z}}^2 \rightarrow R^1 f_* \underline{\mathbb{Z}}$ , associate  $(E, \alpha \circ \gamma)$ . Similarly, viewing  $X$  with  $q : \underline{\mathbb{R}}^2 \otimes \mathcal{O}_X \cong R^1 f_* \underline{\mathbb{R}} \otimes_{\mathbb{R}} \mathcal{O}_X \rightarrow \omega^{-1}$  as classifying complex structures on  $\mathbb{R}^2$ , we see a right action of  $GL_2^+(\mathbb{R})$  on  $(X, \underline{\mathbb{R}}^2, \omega, q)$ .

(2.3) Choose a basis  $(x_1, x_2)$  of  $\mathbb{R}^2$  such that  $x_1 \wedge x_2 > 0$ . A point  $f : \mathbb{R}^2 \rightarrow \mathbb{C}$ , modulo  $\mathbb{C}^\times$ , of  $X$  is parameterized by  $z = f(x_1)/f(x_2)$  ( $\text{Im}(z) > 0$ ), and the map  $q$  identifies with:

$$q : \mathbb{R}^2 \rightarrow \mathbb{G}_a : ax_1 + bx_2 \mapsto az + b.$$

This reveals a non-equivariant trivialization of  $\omega^{-1}$  over  $X$ . Relative to this trivialization, a section  $f(z)$  of  $\omega^k$  on  $X$  is transformed by an element  $\begin{pmatrix} a & c \\ b & d \end{pmatrix}^{-1}$  of  $GL_2^+(\mathbb{R}^2)$  (matrix in the basis  $(x_1, x_2)$ ) into:

$$f \cdot \begin{pmatrix} a & c \\ b & d \end{pmatrix}^{-1} (z) = (cz + d)^{-k} f\left(\frac{az + b}{cz + d}\right).$$

From the identity:

$$dz = (cz + d)^2 d\left(\frac{az + b}{cz + d}\right) \cdot \begin{vmatrix} a & b \\ c & d \end{vmatrix}^{-1},$$

we deduce that  $dz$  is a section of  $\omega^{-2} \otimes \Omega_X^1$  invariant under  $SL_2(\mathbb{R})$ . This section is nowhere vanishing and defines an isomorphism of  $SL_2(\mathbb{R})$ -equivariant sheaves between  $\omega^2$  and  $\Omega_X^1$ .

(2.4) Let  $\Gamma$  be a discrete subgroup of  $SL_2(\mathbb{R})$  with no elements of finite order and with finite volume quotient. It is known that the quotient space  $X/\Gamma$  identifies with a smooth projective curve  $\overline{X/\Gamma}$  minus finitely many points. The group  $\Gamma$  acts without fixed points on  $X$ . The equivariant local system  $\underline{\mathbb{R}}^2$  on  $X$ , along with the equivariant exact sequence:

$$0 \rightarrow \omega \rightarrow \underline{\mathbb{R}}^2 \otimes_{\mathbb{R}} \mathcal{O}_X \xrightarrow{q} \omega^{-1} \rightarrow 0,$$

thus defines on  $X/\Gamma$  a local system  $U$  and an exact sequence:

(2.5)

$$0 \rightarrow \omega \rightarrow U \otimes_{\mathbb{R}} \mathcal{O}_{X/\Gamma} \rightarrow \omega^{-1} \rightarrow 0.$$

In the special case where  $\Gamma \subset SL_2(\mathbb{Z})$ , these structures are derived from the elliptic curve

$E$  on  $X/\Gamma$  whose pullback is the equivariant elliptic curve  $E_X$  on  $X$ .

(2.6) The cusps of  $\overline{X/\Gamma}$  are described as follows (see [9]):

(a) They correspond to conjugacy classes in  $\Gamma$  of non-trivial subgroups of  $\Gamma$ , maximal among subgroups consisting of unipotent elements.

(b) Let  $\Gamma_0 \subset \Gamma$  be such a subgroup, and choose a basis  $(x_1, x_2)$  of  $\mathbb{R}^2$  such that, in this basis,  $\Gamma_0$  is represented by matrices:

$$\begin{pmatrix} 1 & 0 \\ n & 1 \end{pmatrix} \quad (n \in \mathbb{Z}).$$

Let  $z$  be the coordinate (2.3) on  $X$  defined by  $(x_1, x_2)$ . There exists  $N$  such that the region  $X_N = \{z \mid \text{Im}(z) > N\}$  of  $X$  is disjoint from its conjugates under  $\gamma \notin \Gamma_0$ , so that  $X_N/\Gamma_0 \hookrightarrow X/\Gamma$ . The function  $q = e^{2\pi iz}$  establishes an isomorphism between  $X_N/\Gamma_0$  and the punctured disk  $0 < |q| < e^{-2\pi N}$ . If  $P_{\Gamma_0}$  is the cusp of  $\overline{X/\Gamma} - X/\Gamma$  associated to  $\Gamma_0$ , this isomorphism extends to an isomorphism of a neighborhood of  $P_{\Gamma_0}$  with the disk  $0 \leq |q| < e^{-2\pi N}$ .

By (2.3), sections of  $\omega$  over  $X_N$  which are invariant under  $\Gamma_0$  are identified with holomorphic periodic functions of period 1 on  $X_N$ . We still denote by  $\omega$  the invertible sheaf on  $\overline{X/\Gamma}$  extending  $\omega$  such that near a cusp  $P_{\Gamma_0}$ , the section of  $\omega$  over  $X_N/\Gamma_0$  defined by the constant function 1 extends to an invertible section over  $\overline{X_N/\Gamma_0}$ .

(2.7) On  $\overline{X/\Gamma}$ , we have two invertible sheaves  $\Omega^1$  and  $\omega$ , and an isomorphism  $\varphi$  (2.3) between their restrictions to  $X/\Gamma$ . From the formula:

$$dq = de^{2\pi iz} = 2\pi i e^{2\pi iz} dz = 2\pi i q dz,$$

it follows that the map:

$$\varphi : \Omega^1 \rightarrow \omega^2$$

extends to  $\overline{X/\Gamma}$  and has a simple zero at each cusp.

**DEFINITION 2.8.** The space of parabolic automorphic forms of weight  $k + 2$ , relative to  $\Gamma$ , is the space of global sections:

$$H^0(\overline{X/\Gamma}, \Omega^1 \otimes \omega^k).$$

By (2.7), this space also identifies with the space of global sections of  $\omega^{k+2}$  that vanish at the cusps.

(2.9) Let  $U^k$  denote the  $k$ -th symmetric power of the local system  $U$  on  $\overline{X/\Gamma}$ . The map (2.5) induces a map:

$$\iota^k : \omega^k \rightarrow U^k \otimes_{\mathbb{R}} \mathbb{C},$$

and hence a map, still denoted by  $\iota^k$ :

$$\iota^k : \Omega^1 \otimes \omega^k \rightarrow \Omega^1(U^k),$$

where  $\Omega^1(U^k)$  is the sheaf of holomorphic differential forms on  $\overline{X/\Gamma}$  with coefficients in  $U^k$ . The de Rham resolution of  $U^k \otimes_{\mathbb{R}} \mathbb{C}$ :

$$0 \rightarrow U^k \otimes_{\mathbb{R}} \mathbb{C} \rightarrow U^k \otimes_{\mathbb{R}} \mathcal{O}_{X/\Gamma} \xrightarrow{d} U^k \otimes_{\mathbb{R}} \Omega^1 \rightarrow 0$$

induces a map:

$$\delta : H^0(X/\Gamma, \Omega^1(U^k)) \rightarrow H^1(X/\Gamma, U^k \otimes \mathbb{C}).$$

Furthermore, the cohomology space  $H^1(X/\Gamma, U^k \otimes \mathbb{C})$  has a natural complex conjugation, so  $\delta$  defines a conjugate-linear map  $\bar{\delta}$  from the complex conjugate space of  $H^0(X/\Gamma, \Omega^1(U^k))$  to  $H^1(X/\Gamma, U^k \otimes \mathbb{C})$ . This gives a map  $sh_0 = \delta \cdot H^0(\iota^k) \oplus \bar{\delta} \cdot H^0(\iota^k)$ :

$$sh_0 : H^0(X/\Gamma, \Omega^1 \otimes \omega^k) \oplus \overline{H^0(X/\Gamma, \Omega^1 \otimes \omega^k)} \rightarrow H^1(X/\Gamma, U^k \otimes \mathbb{C}).$$

For any sheaf  $F$  on a space  $Y$ , denote by  $\tilde{H}^i(Y, F)$  the image of the compactly supported cohomology  $H_c^i(Y, F)$  in the ordinary cohomology  $H^i(Y, F)$ .

Theorem 4.2.6 of [9] is essentially equivalent to the following theorem (in loc. cit.,  $k$  is assumed even, but the same proof works in general):

**THEOREM 2.10 (Shimura [7]).** There exists an isomorphism  $sh$  making the following diagram commute:

$$\begin{array}{ccc} H^0(\overline{X/\Gamma}, \Omega^1 \otimes \omega^k) \oplus H^0(\overline{X/\Gamma}, \Omega^1 \otimes \omega^k) & \xrightarrow{sh} & \tilde{H}^1(X/\Gamma, U^k \otimes \mathbb{C}) \\ \downarrow & & \downarrow \\ H^0(X/\Gamma, \Omega^1 \otimes \omega^k) \oplus \overline{H^0(X/\Gamma, \Omega^1 \otimes \omega^k)} & \xrightarrow{sh_0} & H^1(X/\Gamma, U^k \otimes \mathbb{C}) \end{array}$$

We call  $sh$  the *Shimura isomorphism*.

(2.11) In the special case where  $\Gamma$  is a finite-index subgroup of  $SL_2(\mathbb{Z})$ , the elliptic curve  $E$  on  $X/\Gamma$  comes from a scheme of elliptic curves over the algebraic curve  $X/\Gamma$  (i.e., its modular invariant is meromorphic at infinity); it thus admits a Néron model  $\bar{E}$  over  $\overline{X/\Gamma}$ . One can show that the fibers of  $\bar{E}$  at the cusps are of multiplicative type, and that over the entire  $\overline{X/\Gamma}$ , we have  $\omega = e^* \Omega_{\bar{E}/(\overline{X/\Gamma})}^1$ .

In this case,  $U = R^1 f_* \mathbb{Z} \otimes \mathbb{R}$ , so the target of the Shimura isomorphism rewrites:

$$\tilde{H}^1(X/\Gamma, U^k \otimes \mathbb{C}) \cong \tilde{H}^1(X/\Gamma, \text{Sym}^k(R^1 f_* \mathbb{Z})) \otimes_{\mathbb{Z}} \mathbb{C}.$$

## Chapter 3

# Hecke operators and fundamental $\ell$ -adic representations

### No.3 - Hecke Operators and the Fundamental $\ell$ -adic Representation.

(3.1) Recall (cf. [3]) that the category of "locally constant" constructible  $\mathbb{Z}_\ell$ -sheaves (abbreviated as l.c.c.) on a scheme  $S$  is the category of projective systems of sheaves  $\{F_n\}_{n \in \mathbb{N}}$  on the étale site  $S_{\text{ét}}$  of  $S$  satisfying:

- (i)  $F_n$  is a locally constant sheaf of  $\mathbb{Z}/(\ell^n)$ -modules of finite type;
- (ii) If  $n \leq m$ , then  $F_m \otimes \mathbb{Z}/(\ell^n) \xrightarrow{\sim} F_n$ .

The l.c.c.  $\mathbb{Z}_\ell$ -sheaves form a stack in abelian categories over  $S$ ; the stack of l.c.c.  $\mathbb{Q}_\ell$ -sheaves is the quotient of this stack by the thick sub-stack of l.c.c.  $\mathbb{Z}_\ell$ -sheaves annihilated by a power of  $\ell$ . We denote by  $\otimes_{\mathbb{Q}_\ell}$  the canonical functor from the category of l.c.c.  $\mathbb{Z}_\ell$ -sheaves to that of l.c.c.  $\mathbb{Q}_\ell$ -sheaves.

If  $S$  is connected with a geometric point  $s$ , the category of l.c.c.  $\mathbb{Z}_\ell$ -sheaves (resp.  $\mathbb{Q}_\ell$ -sheaves) on  $S$  is equivalent, via the "Fiber at  $s$ " functor, to the category of continuous representations of the fundamental group  $\pi_1(S, s)$  on a finite-type  $\mathbb{Z}_\ell$ -module (resp. a finite-rank  $\mathbb{Q}_\ell$ -vector space).

For a finite set  $T$  of primes, an l.c.c.  $A^T$ -sheaf consists of data: for each prime  $\ell$ , a l.c.c.  $\mathbb{Z}_\ell$ -sheaf if  $\ell \notin T$ , and a l.c.c.  $\mathbb{Q}_\ell$ -sheaf if  $\ell \in T$ . For  $T = \emptyset$ , we speak of l.c.c.  $\mathbb{Z}_\ell$ -sheaves rather than l.c.c.  $A^T$ -sheaves.

For arbitrary  $T$ , the category of l.c.c.  $A^T$ -sheaves is the inductive limit of categories of l.c.c.  $A^{T'}$ -sheaves for finite  $T' \subset T$ . We set:

$$\begin{aligned} \mathbb{Z}_\ell &= \varprojlim \mathbb{Z}/(\ell^n), \quad \mathbb{Q}_\ell = \mathbb{Z}_\ell \otimes \mathbb{Q}, \\ \widehat{\mathbb{Z}} &= \prod_{\ell} \mathbb{Z}_\ell \quad \text{and} \quad \mathbb{A}_T^f = \widehat{\mathbb{Z}} \otimes \mathbb{A}_T^f. \end{aligned}$$

The stack of elliptic curves up to isogeny over  $S$  is obtained by formally inverting isogenies in the stack of elliptic curves over  $S$ . Denote by  $\otimes_{\mathbb{Q}}$  the functor associating to an elliptic curve

its underlying isogeny class. For  $S$  quasi-compact, we have

$$\mathrm{Hom}(E, F) \otimes \mathbb{Q} \simeq \mathrm{Hom}(E \otimes \mathbb{Q}, F \otimes \mathbb{Q}),$$

and for  $S$  normal, every elliptic curve up to isogeny over  $S$  underlies an elliptic curve over  $S$ .

(3.2) Let  $f : E \rightarrow S$  be an elliptic curve over a scheme  $S$ . Define  $T_\ell(E)$  as the projective system of kernels  $E[\ell^n]$  of multiplication by  $\ell^n$  in  $E$ , with transition maps  $E[\ell^n] \rightarrow E[\ell^m]$  ( $n \geq m$ ) given by multiplication by  $\ell^{n-m}$ . Similarly for  $\mathbb{G}_m$ , set  $T_\ell(\mathbb{G}_m) = \mathbb{Z}_\ell(1)$ . If  $\ell$  is invertible on  $S$ ,  $T_\ell(E)$  and  $\mathbb{Z}_\ell(1)$  are  $\mathbb{Z}_\ell$ -sheaves on  $S$ . Define  $T_\infty(E)$  as the relative Lie algebra of  $E$  over  $S$  (the invertible sheaf dual to  $\omega$  in (2.1(a))).

Assume  $S$  has characteristic 0. Define the  $\widehat{\mathbb{Z}}$ -sheaf  $T_f(E)$  on  $S$  as the system of  $T_\ell(E)$ , and set  $V_f(E) = T_f(E) \otimes \mathbb{A}^f$ . For an isogeny  $u : E \rightarrow F$ ,  $u$  induces isomorphisms  $V_f(E) \rightarrow V_f(F)$  and  $T_\infty(E) \rightarrow T_\infty(F)$ ; thus the functors  $V_f$  and  $T_\infty$  factor through the category of elliptic curves up to isogeny over  $S$ .

**PROPOSITION 3.3.** Let  $S$  be a scheme of characteristic 0,  $\underline{E}_1(S)$  the category of elliptic curves over  $S$ , and  $\underline{E}_2(S)$  the category of triples: an elliptic curve up to isogeny  $E$  over  $S$ , a  $\widehat{\mathbb{Z}}$ -sheaf  $T$  isomorphic to  $\widehat{\mathbb{Z}}^2$ , and an isomorphism  $\beta : V_f(E) \simeq T \otimes \mathbb{A}$ . The functor  $I : E \mapsto (E \otimes \mathbb{Q}, T_f(E), V_f(E) \sim T_f(E) \otimes \mathbb{A})$  from  $\underline{E}_1(S)$  to  $\underline{E}_2(S)$  is an equivalence of categories.

The question is local on  $S$ , which we may assume to be quasi-compact. If  $f : E \rightarrow F$  is a morphism of elliptic curves over  $S$ , and if  $f$  is an isogeny, we have an exact sequence (3.4):

$$0 \rightarrow T_f(E) \rightarrow T_f(F) \rightarrow \mathrm{Ker}(f) \rightarrow 0.$$

A morphism  $f$  is divisible by  $n$  iff it annihilates the kernel  $E[n]$ , since multiplication by  $n$  on  $E/E[n]$  is an isomorphism. By (3.4), this occurs iff  $T_f(f)$  is divisible by  $n$ , showing that  $\mathrm{Hom}_S(E, F)$  is the subgroup of  $\mathrm{Hom}_S(E \otimes \mathbb{Q}, F \otimes \mathbb{Q})$  consisting of morphisms  $f$  where  $V_f(f)$  maps  $T_f(E)$  into  $T_f(F)$ . Thus  $I$  is fully faithful.

Let  $X \in \mathrm{Ob}(\underline{E}_2(S))$ . Locally on  $S$ ,  $X$  is defined by an elliptic curve up to isogeny  $E \otimes \mathbb{Q}$  and a "lattice"  $T$  in  $V_f(E)$  coinciding with  $T_\ell(E)$  for almost all  $\ell$ . For  $q \in \mathbb{Q}$ ,  $(E \otimes \mathbb{Q}, T)$  is isomorphic to  $(E \otimes \mathbb{Q}, qT)$ , allowing us to assume  $T_f(E) \subset T$ .

The quotient  $K = T/T_f(E)$  is canonically isomorphic to a finite subgroup of  $E$ , and  $X$  is the image under  $I$  of  $E/K$  (cf. 3.4).  $\square$

**COROLLARY 3.5.** The functor  $F_1$  (resp.  $F'_1$ ) associating to each scheme  $S$  of characteristic 0 the set of isomorphism classes of elliptic curves  $E$  over  $S$  equipped with an isomorphism  $\alpha : T_f(E) \xrightarrow{\sim} \widehat{\mathbb{Z}}^2$  (resp. and an isomorphism  $\alpha_\infty : T_\infty(E) \xrightarrow{\sim} \mathbb{G}_a$ ) is isomorphic to the functor  $F_2$  (resp.  $F'_2$ ) associating to  $S$  the set of isomorphism classes of elliptic curves up to isogeny  $F$  over  $S$  equipped with an isomorphism  $\beta : V_f(F) \xrightarrow{\sim} (\mathbb{A}^f)^2$  (resp. and an isomorphism  $\beta_\infty : T_\infty(F) \xrightarrow{\sim} \mathbb{G}_a$ ).

**PROPOSITION 3.6.** The functor  $F_1$  (resp.  $F'_1$ ) is represented by a scheme  $\mathcal{M}_\infty$  (resp.  $\mathcal{M}'_\infty$ ) over  $\mathbb{Q}$ .

Let  $n \geq 3$ . The functor associating to each scheme  $S$  the set of isomorphism classes of elliptic curves equipped with an isomorphism  $\alpha_n : E[n] \xrightarrow{\sim} (\mathbb{Z}/n)^2$  (resp. and  $\alpha_\infty : T_\infty(E) \xrightarrow{\sim} \mathcal{O}_S$ ) is represented by an affine curve  $\mathcal{M}_n$  (resp. an affine surface  $\mathcal{M}'_n$ ) over  $\mathrm{Spec}(\mathbb{Z}[1/n])$ . For  $n|m$ ,

the morphism  $\mathcal{M}_m \rightarrow \mathcal{M}_n$  defined by

$$(E, \alpha_m : E[m] \xrightarrow{\sim} (\mathbb{Z}/m)^2) \mapsto (E, \frac{n}{m}\alpha_m : E[n] \xrightarrow{\sim} (\mathbb{Z}/n)^2)$$

is finite étale over  $\text{Spec}(\mathbb{Z}[1/m])$ , and we have

$$\mathcal{M}_\infty = \varprojlim_n \mathcal{M}_n.$$

The same procedure applies to represent  $F'_1$ .

(3.7) The scheme  $\mathcal{M}_\infty$  (resp.  $\mathcal{M}'_\infty$ ) carries a universal elliptic curve  $f_\infty : \mathcal{E} \rightarrow \mathcal{M}_\infty$  (resp.  $f'_\infty : \mathcal{E}_\infty \rightarrow \mathcal{M}'_\infty$ ) and an isomorphism  $\alpha : T_f(\mathcal{E}) \xrightarrow{\sim} \hat{\mathbb{Z}}^2$  (resp. and  $\alpha_\infty : T_\infty(\mathcal{E}_\infty) \xrightarrow{\sim} \mathbb{G}_a$ ).

By (3.5),  $\mathcal{M}_\infty$  represents  $F_2$  (resp.  $F'_2$ ), which highlights a left action of the adelic group  $\text{GL}_2(\mathbb{A}^f)$  on  $(\mathcal{M}_\infty, \mathcal{E}_\infty \otimes \mathbb{Q}, \alpha \otimes (\mathbb{A}^f)^2)$  (resp.  $(\mathcal{M}_\infty, \mathcal{E}_\infty \otimes \mathbb{Q}, \alpha \otimes (\mathbb{A}^f)^2, \alpha_\infty)$ ), given on the functor, for  $g \in \text{GL}_2(\mathbb{A}^f)$  by:

$$g : (F, \beta : V_f(E) \xrightarrow{\sim} (\mathbb{A}^f)^2, \beta_\infty) \mapsto (F, g \circ \beta : V_f(E) \xrightarrow{\sim} (\mathbb{A}^f)^2, \beta_\infty).$$

Šafarevič first noted this fact.

Let  $Y$  be a scheme over  $\mathbb{C}$ , which is a projective limit of finite-type schemes  $Y_i$  over  $\mathbb{C}$  with finite transition maps. The locally ringed space  $Y^{\text{an}}$ , as the projective limit of  $Y_i^{\text{an}}$ , depends only on  $Y$  and not on its representation as a projective limit. If  $Y$  is a scheme over  $\mathbb{Q}$ , which is a projective limit of finite-type schemes  $Y_i$  over  $\mathbb{Q}$  with finite transition maps, set  $Y^{\text{an}} = (Y \otimes \mathbb{C})^{\text{an}}$ . This applies to  $\mathcal{M}_\infty$  and  $\mathcal{M}'_\infty$ .

**PROPOSITION 3.8.** We have canonical isomorphisms:

$$\mathcal{M}'_\infty{}^{\text{an}} \simeq \text{Isom}(\mathbb{Q}^2 \otimes \mathbb{A}, \mathbb{C} \times (\mathbb{A}^f)^2) / \text{GL}_2(\mathbb{Q})$$

$$\mathcal{M}_\infty{}^{\text{an}} \simeq \mathbb{C}^* \backslash \text{Isom}(\mathbb{Q}^2 \otimes \mathbb{A}, \mathbb{C} \times (\mathbb{A}^f)^2) / \text{GL}_2(\mathbb{Q}),$$

or less canonically:

$$\mathcal{M}'_\infty{}^{\text{an}} \simeq \text{GL}_2(\mathbb{A}) / \text{GL}_2(\mathbb{Q}),$$

$$\mathcal{M}_\infty{}^{\text{an}} \simeq K_\infty \backslash \text{GL}_2(\mathbb{A}) / \text{GL}_2(\mathbb{Q}),$$

where  $K_\infty$  is the maximal compact subgroup at infinity plus real homotheties. These isomorphisms respect the  $\text{GL}_2(\mathbb{A}^f)$ -action.

The notion of elliptic curves up to isogeny extends to complex analytic geometry. An isogeny  $\varphi : E \rightarrow F$  induces an isomorphism  $\varphi^*$  between rational cohomology local systems, which allows to define the latter for a curve up to isogeny. For a complex analytic space  $S$ , using (2.1), giving an elliptic curve up to isogeny over  $S$  is equivalent to giving: an invertible sheaf  $T_\infty$ , a local system  $T_\mathbb{Q}$  of  $\mathbb{Q}$ -vector spaces, and a morphism  $u : T_\mathbb{Q} \rightarrow T_\infty$  inducing pointwise isomorphisms between  $T_\mathbb{Q} \otimes \mathbb{R}$  and  $T_\infty$ .

Let  $n$  be an integer and  $K_n$  the kernel of the natural map  $\prod \text{GL}_2(\mathbb{Z}_\ell) \rightarrow \text{GL}_2(\mathbb{Z}/n)$ .

Let  $G_1$  be the functor associating to  $S$  the set of isomorphism classes of elliptic curves  $f : E \rightarrow S$  over  $S$  equipped with isomorphisms  $\varphi : \mathbb{Q}^2 \xrightarrow{\sim} T_\mathbb{Q}(E)$ ,  $\alpha_\infty : T_\infty(E) \xrightarrow{\sim} \mathcal{O}_S$ , and  $\alpha_n : E[n] \xrightarrow{\sim} (\mathbb{Z}/n)^2$ . As in (3.3),  $G_1$  is isomorphic to the functor  $G_2$  associating to  $S$  isogeny classes of elliptic curves  $E$  over  $S$  with  $\varphi : \mathbb{Q}^2 \xrightarrow{\sim} T_\mathbb{Q}(E)$ ,  $\alpha_\infty : T_\infty(E) \xrightarrow{\sim} \mathcal{O}_S$ , and an

isomorphism  $V_f(E) \xrightarrow{\sim} (\mathbb{A}^f)^2$  given locally over  $S$  up to the composition by an element of  $K_n$ . Such objects are determined by a composite map  $\varphi'$  (defined locally modulo  $K_n$ ):

$$\varphi' : \mathbb{Q}^2 \xrightarrow{\varphi} T_{\mathbb{Q}}(E) \rightarrow T_{\infty}(E) \times V_f(E) \xrightarrow{\sim} \mathcal{O}_S \times (\mathbb{A}^f)^2,$$

we have

$$E = \mathcal{O}_S / \varphi'(\mathbb{Q}^2 \cap \varphi'^{-1}(T_{\infty}(E) \times T_f(E))) = \widehat{\mathbb{Z}}^2 \backslash \mathcal{O}_S \times (\mathbb{A}^f)^2 / \varphi'(\mathbb{Q}^2),$$

so that (cf. 2.2)  $G_1$  and  $G_2$  are represented by

$$K_n \backslash \text{Isom}(\mathbb{Q}^2 \otimes \mathbb{A}, \mathbb{C} \times (\mathbb{A}^f)^2).$$

Assuming now  $n \geq 3$ , so that  $\text{GL}_2(\mathbb{Q})$  acts freely on the previous space. The analytic space  $\mathcal{M}_n^{\text{an}}$  (resp.  $\mathcal{M}_n'^{\text{an}}$ ) represents the analogous functor, in analytical geometry, of the functor that is represented by  $\mathcal{M}_n$  (resp.  $\mathcal{M}_n'$ ) because this functor  $X$ , is representable and the map  $X \rightarrow \mathcal{M}_n^{\text{an}}$  (resp.  $X \rightarrow \mathcal{M}_n'^{\text{an}}$ ) induces a bijection on the set of points with values in any finite rank  $\mathbb{C}$ -algebra.

Thus we get:

$$\mathcal{M}_n'^{\text{an}} \simeq K_n \backslash \text{Isom}(\mathbb{Q}^2 \otimes \mathbb{A}, \mathbb{C} \times (\mathbb{A}^f)^2) / \text{GL}_2(\mathbb{Q}).$$

Similarly for  $\mathcal{M}_n$ , we obtain the first claim in (3.8) via passing to the projective limit of  $n$ .

A point  $x$  in  $\text{Isom}(\mathbb{Q}^2 \otimes \mathbb{A}, \mathbb{C} \times (\mathbb{A}^f)^2) / \text{GL}_2(\mathbb{Q})$  corresponds to a "lattice"  $L_x \subset \mathbb{C} \times (\mathbb{A}^f)^2$ , and the curve corresponding to  $x$  is:

$$E_x \sim \widehat{\mathbb{Z}}^2 \backslash \mathbb{C} \times (\mathbb{A}^f)^2 / L_x,$$

equipped with  $V_f(E_x) \simeq L_x \otimes \mathbb{A}^f \simeq (\mathbb{A}^f)^2$ . This easily yields the last claim in (3.8).  $\square$

Let  $f_n : \mathcal{E} \rightarrow \mathcal{M}_n$  be the universal elliptic curve over  $\mathcal{M}_n$ . Fixing an integer  $k$ , we define:

**DEFINITION 3.9.** Let  $W$  (or  ${}^k W$  if there is risk of ambiguity) be the  $\mathbb{Q}$ -vector space:

$$W = \varinjlim_n \tilde{H}^1(\mathcal{M}_n^{\text{an}}, \text{Sym}^k(R^1 f_{n*} \mathbb{Q})) = \varinjlim_n W.$$

This vector space does not depend on the universal elliptic curve (up to isogeny)  $f_{\infty} : \mathcal{E} \rightarrow M_{\infty}$  so that, by transport of structure, it is endowed with a left action of  $\text{GL}_2(\mathbb{A}^f)$ .

For a prime  $\ell$ , the  $\mathbb{Q}_{\ell}$ -vector space  $W_{\ell} = W \otimes \mathbb{Q}_{\ell}$  admits an algebraic definition via  $\ell$ -adic cohomology over  $\overline{\mathbb{Q}}$  deduced from extension of scalar of  $\mathcal{M}_n$ :

$$(3.10) \quad W_{\ell} = \varinjlim_n \tilde{H}^1(\mathcal{M}_n \otimes \overline{\mathbb{Q}}, \text{Sym}^k(R^1 f_{n*} \mathbb{Q}_{\ell})) = \varinjlim_n W_{\ell},$$

endowed with a Galois action on  $W_{\ell}$  and  ${}_n W_{\ell}$ .

Finally, the space  $\mathcal{M}_n^{\text{an}}$  is a disjoint union of quotients of the Poincaré upper half-plane by congruence subgroups of  $\text{SL}_2(\mathbb{Z})$ . Let  $\omega$  be the invertible sheaf on  $\mathcal{M}_n$  defined by  $\mathcal{E}$ . Shimura's theory (2.10) gives:

$$(3.11) \quad W_{\infty} = W \otimes \mathbb{C} = \varinjlim_n \left( H^0(\overline{\mathcal{M}}_n^{\text{an}}, \Omega^1 \otimes \omega^k) \oplus H^0(\overline{\mathcal{M}}_n^{\text{an}}, \overline{\Omega}^1 \otimes \omega^k) \right).$$

This decomposition of  $W \otimes \mathbb{C}$  into two complex conjugate subspaces- one being the space of holomorphic parabolic modular forms of weight  $k+2$ -resembles a Hodge decomposition of type  $(0, k+1) + (k+1, 0)$ .

The adelic action commutes with the Galois action and preserves this decomposition.

Though the  $\ell$ -adic local system  $R^1 f_{n*} \underline{\mathbb{Q}}_\ell$  is trivial on  $\mathcal{M}_\infty$ , I do not know if  $W_\ell$  relates to  $\varinjlim_n \left( \tilde{H}^1(\mathcal{M}_n \otimes \overline{\mathbb{Q}}, \underline{\mathbb{Q}}_\ell) \otimes \text{Sym}^k(\mathbb{Q}_\ell^2) \right)$ .

(3.12) For  $n \geq 3$  and  $K_n$  as in (3.8), we have  $W^{K_n} = {}_n W$ . This is verified by passing to the limit, and results from the fact that in rational cohomology, the cohomology of a quotient of a space by a finite group is obtained by taking the invariants of this group in the cohomology.

Let, for  $p$  prime,  $W^{(p)} = W^{\text{GL}_2(\mathbb{Z}_p)}$ . By passing to the limit, we get

$$W^{(p)} = \varinjlim_{(n,p)=1} {}_n W.$$

This cohomology space carries actions by:

- (i) The subgroup  $\prod_{\ell \neq p} \text{GL}_2(\mathbb{Q}_\ell) \subset \text{GL}_2(\mathbb{A}^f)$ , centralizing  $\text{GL}_2(\mathbb{Z}_p)$ ;
- (ii) The Hecke algebra  $\underline{H}(\text{GL}_2(\mathbb{Q}_p), \text{GL}_2(\mathbb{Z}_p))$ , algebra of integer measures on the discrete space  $\text{GL}_2(\mathbb{Q}_p)/\text{GL}_2(\mathbb{Z}_p)$  left-invariant under action by  $\text{GL}_2(\mathbb{Z}_p)$ : This sub-algebra of the group algebra  $\text{GL}_2(\mathbb{Q}_p)$  acts on  $W$  in accordance with  $W^{(p)}$ . This algebra already acts on each  ${}_n W$  for every  $n$  prime to  $p$ .

The Hecke algebra has a basis of (measures associated to characteristic functions) double cosets of  $\text{GL}(\mathbb{Z}_p)$  in  $\text{GL}(\mathbb{Q}_p)$ , and we know that

$$\underline{H}(\text{GL}(\mathbb{Q}_p), \text{GL}(\mathbb{Z}_p)) = \mathbb{Z}[T_p, R_p, R_p^{-1}],$$

where  $T_p$  and  $R_p$  correspond to the double cosets of  $\begin{pmatrix} 1 & 0 \\ 0 & p^{-1} \end{pmatrix}$  and  $\begin{pmatrix} p^{-1} & 0 \\ 0 & p^{-1} \end{pmatrix}$ .

(3.13) For a prime  $p$ , integer  $n \geq 3$  coprime to  $p$ , define  $F_{n,p}$  as the functor that associates to a scheme  $S$  the set of isomorphism classes of commutative diagrams of  $S$ -schemes:

$$(3.14) \quad \begin{array}{ccc} & (\underline{\mathbb{Z}}/n)^2 & \\ \alpha \nearrow & & \nwarrow \alpha' \\ E[n] & \xrightarrow{\quad} & F[n] \\ \downarrow & & \downarrow \\ E & \xrightarrow{\quad \varphi \quad} & F \end{array}$$

where  $\varphi$  is a  $p$ -isogeny between elliptic curves and  $\alpha$  is an isomorphisms. Let  $q_1, q_2 : F_{n,p} \rightarrow \mathcal{M}_n$  be the morphism of functors that associates to a diagram (3.14) the subdiagram  $(E, E_n, \alpha)$ , or  $(F, F_n, \alpha')$ .

**PROPOSITION 3.15.** The functor  $F_{n,p}$  is represented by a scheme  $\mathcal{M}_{n,p}$ , with  $q_1, q_2 : \mathcal{M}_{n,p} \rightarrow \mathcal{M}_n$  finite.

The automorphism  $\sigma$  of  $F_{n,p}$  swapping  $\varphi : E \rightarrow F$  and  ${}^t\varphi : F \rightarrow E$  exchanges  $q_1$  and  $q_2$ . It suffices then to consider  $q_1$ . This morphism identifies  $F_{n,p}$  with the functor of subgroups of order  $p$  of the universal elliptic curve  $\mathcal{E}$  over  $\mathcal{M}_n$ , such that, by the theory of Hilbert schemes,  $F_{n,p}$  is representable and  $\mathcal{M}_{n,p}$  is proper over  $\mathcal{M}_n$ . If  $s$  is a geometric point of  $\mathcal{M}_n$ ,  $q_1^{-1}(s)$  is

the set of subgroups of order  $p$  of  $E_s$ , and has  $p + 1$  elements if  $\text{char}(k(s)) \neq p$ , has one element (the kernel of Frobenius) if  $\text{char}(k(s)) = p$ .  $\square$

One can show that  $M_{n,p}$  is regular, and that  $q_1$  and  $q_2$  are finite and flat; We do not use this delicate result, contenting ourselves here to note that over  $\text{Spec}(\mathbb{Z}[1/p])$ , each  $q_i$  becomes étale of degree  $p + 1$  on  $\mathcal{M}_n$ .

These morphisms  $q_i (i = 1, 2)$  fit into a commutative diagram:

$$(3.16) \quad \begin{array}{ccccc} & q_1^* \mathcal{E} & \xrightarrow{\varphi} & q_2^* \mathcal{E} & \\ & \swarrow u & & \searrow v & \\ \mathcal{E} & & & & \mathcal{E} \\ & \searrow f_n & & \swarrow f_n & \\ & \mathcal{M}_n & & \mathcal{M}_n & \end{array}$$

where  $(\varphi, u, v)$  is a part of universal diagram (3.14).

Let  $I_p$  denote the endomorphism of  $\mathcal{M}_n$  induced by  $(E, \alpha) \mapsto (E, \alpha/p)$ :

$$(3.17) \quad \begin{array}{ccc} \mathcal{E} & \xrightarrow{\quad} & \mathcal{E} \\ \downarrow & & \downarrow \\ \mathcal{M}_n & \xrightarrow{I_p} & \mathcal{M}_n \end{array}$$

$I_p^*$  is an automorphism of  $\tilde{H}^i(\mathcal{M}_n^{\text{an}}, \text{Sym}^k(R^1 f_{n*} \underline{\mathbb{Z}}))$ .

It is tedious but routine to show that

**PROPOSITION 3.18.**

(i) The Hecke operator  $T_p$  on  $W_n$  is expressed, with the help of (3.16), as the composite map

$$\begin{aligned} \tilde{H}^1(\mathcal{M}_n^{\text{an}}, \text{Sym}^k(R^1 f_{n*} \underline{\mathbb{Q}})) &\xrightarrow{q_2^*} \tilde{H}^1(\mathcal{M}_{n,p}^{\text{an}}, \text{Sym}^k(R^1 v_* \underline{\mathbb{Q}})) \\ &\xrightarrow{\varphi^*} \tilde{H}^1(\mathcal{M}_n^{\text{an}}, \text{Sym}^k(R^1 u_* \underline{\mathbb{Q}})) \xrightarrow{q_{1*}} \tilde{H}^1(\mathcal{M}_n^{\text{an}}, \text{Sym}^k(R^1 f_{n*} \underline{\mathbb{Q}})) \end{aligned}$$

where  $q_{1*}$  is the "trace map" for the covering  $q_1$ .

(ii) Similarly,  $R_p = p^k I_p^*$ .  $\square$

The suspicious reader might forget about the adelic preliminaries and define  $T_p$  by (i).

For  $n = 1$  or  $2$ , set  ${}_n W = W^{K_n}$ , so that

$${}_1 W = {}_n W^{\text{GL}_2(\mathbb{Z}/n\mathbb{Z})}.$$

Let  $S_{k+2}$  denote the space of parabolic modular forms of weight  $k + 2$  for  $\text{SL}_2(\mathbb{Z})$ . Shimura's isomorphism (3.11) gives:

$${}_1^k W_\infty = {}_1^k W \otimes \mathbb{C} = S_{k+2} \oplus \overline{S_{k+2}}.$$

It is tedious but routine to show that

**PROPOSITION 3.19.-** The Hecke operator  $T_p$  on  ${}^k_1W_\infty$  corresponds under Shimura's isomorphism to the direct sum of  $T_p$  on  $S_{k+2}$  (including  $p^{k-1}$  factor) and its conjugate.  $\square$

(3.20) We have canonically:

$$\Lambda^2 R^1 f_{n*} \underline{\mathbb{Z}}_\ell \simeq R^2 f_{n*} \underline{\mathbb{Z}}_\ell \simeq \underline{\mathbb{Z}}_\ell(-1),$$

endowing  $\text{Sym}^k(R^1 f_{n*} \underline{\mathbb{Z}}_\ell)$  with a bilinear form (symmetric/alternating for  $k$  even/odd) valued in  $\underline{\mathbb{Z}}_\ell(-k)$ . The form that is induced by tensoring with  $\mathbb{Q}_\ell$  is nondegenerate.

If  $\underline{F}$  is a l.c.c  $\mathbb{Q}_\ell$  sheaf on a scheme  $X$  smooth purely of dimension  $n$  on an algebraically closed field  $k$ , then the Poincaré duality gives

$$\begin{aligned} H^i(X, \underline{F})^\vee &\simeq H_c^{2n-i}(X, \underline{\text{Hom}}(\underline{F}, \underline{\mathbb{Q}}_\ell(n))) \\ H_c^i(X, \underline{F})^\vee &\simeq H^{2n-i}(X, \underline{\text{Hom}}(\underline{F}, \underline{\mathbb{Q}}_\ell(n))) \\ \text{from which } \tilde{H}^i(X, \underline{F})^\vee &\simeq \tilde{H}^{2n-i}(X, \underline{\text{Hom}}(\underline{F}, \underline{\mathbb{Q}}_\ell(n))). \end{aligned}$$

Take  $X = \overline{\mathcal{M}}_n$  and  $\underline{F} = \text{Sym}^k(R^1 f_{n*} \underline{\mathbb{Q}}_\ell)$  into consideration, we define a nondegenerate bilinear form  ${}_n(\ , \ )$  on  ${}^k_nW_\ell$  with value in  $\mathbb{Q}_\ell(-k-1)$ . This form is symmetric for odd  $k$ , alternative for even  $k$ . This is the  $\ell$ -adic analogue of Petersson scalar product. For  $n|m$  with covering  $\psi : \mathcal{M}_m \rightarrow \mathcal{M}_n$  of degree  $d$ , we have:

$${}_m(\psi^*x, \psi^*y) = d \cdot {}_n(x, y).$$

## Chapter 4

# The congruence formula

### No. 4 – The Congruence Formula.

We fix in this section integers  $k \geq 0$  and  $n \geq 3$ , and prime numbers  $p$  and  $\ell$ . We assume that  $p$  is prime to  $n$  and to  $\ell$ . Let  $f : \mathcal{E} \rightarrow \mathcal{M}_n$  be the universal elliptic curve on  $\mathcal{M}_n$ , equipped with  $\alpha : \mathcal{E}[n] \simeq (\mathbb{Z}/n\mathbb{Z})^2$ .

For any scheme  $Y$ , we denote by  $a$  the unique morphism from  $Y$  to  $\text{Spec}(\mathbb{Z})$ , or, if appropriate, to a subscheme of  $\text{Spec}(\mathbb{Z})$ . If  $Y$  is separated and of finite type over  $\text{Spec}(\mathbb{Z})$  and if  $\underline{F}$  is a  $\mathbb{Z}_\ell$ - or  $\mathbb{Q}_\ell$ -sheaf on  $Y$ , we denote by  $R^i a_*(Y, \underline{F})$  (resp.  $R^i a_!(Y, \underline{F})$ ) the  $i$ th direct image (resp. the  $i$ th direct image with proper support, resp.  $\text{Im}(R^i a_!(Y, \underline{F}) \rightarrow R^i a_*(Y, \underline{F}))$ ) of  $\underline{F}$  by  $a$ .

We set, for an integer  $m$ ,  $Y[m] = Y \times \text{Spec}(\mathbb{Z}[1/m])$ .

**THEOREM 4.1 (Igusa [1]).** - The scheme  $\mathcal{M}_n$  can be compactified into a curve scheme  $\mathcal{M}_n^*$ , projective and smooth over  $\text{Spec}(\mathbb{Z}[1/n])$ , such that  $\mathcal{M}_n^* \setminus \mathcal{M}_n$  is an étale cover of  $\text{Spec}(\mathbb{Z}[1/n])$ .

The scheme  $\mathcal{M}_n$  is formally smooth, thus smooth over  $\text{Spec}(\mathbb{Z})$ .

The modular invariant  $j$  of the universal curve on  $\mathcal{M}_n$  defines a morphism

$$j : \mathcal{M}_n \longrightarrow \mathbb{A}_{\text{Spec}(\mathbb{Z}[1/n])}^1.$$

over  $\text{Spec}(\mathbb{Z}[1/n])$ .

This morphism  $j$  is finite and is an étale covering outside the sections 0 and 1728 of  $\mathbb{A}^1$ ; indeed:

- (a) Two elliptic curves over an algebraically closed field with the same  $j$ -invariant are isomorphic (e.g., [8] 6.3), so that the geometric fibers of  $j$  are finite. Since the schemes  $\mathcal{M}_n$  and  $\mathbb{A}^1$  are smooth of the same relative dimension over  $\text{Spec}(\mathbb{Z})$ ,  $j$  is quasi-finite and flat.
- (b) If  $E$  is an elliptic curve over the field of fractions  $K$  of a discrete valuation ring  $R$ , with  $j \in R$  and whose  $n$ -torsion points are rational over  $K$ , then  $E$  has good reduction. The valuative criterion of properness then shows that  $j$  is proper.
- (c) If  $E$  and  $F$  are two elliptic curves over a scheme  $S$  with the same  $j$ -invariant, and if  $j$  and  $j - 1728$  are invertible, then the scheme  $\text{Isom}(S; E, F)$  of isomorphisms between  $E$

and  $F$  is étale over  $S$  (see [8] 6.3). In the diagram

$$\begin{array}{ccc} \underline{\text{Isom}}(\mathcal{M}_n \times_{\mathbb{A}^1} \mathcal{M}_n; \text{pr}_1^* \mathcal{E}, \text{pr}_2^* \mathcal{E}) & \xrightarrow{\cong} & \mathcal{M}_n \times \text{GL}_2(\mathbb{Z}/n) \\ \downarrow & & \downarrow \\ \mathcal{M}_n \times_{\mathbb{A}^1} \mathcal{M}_n & \xrightarrow{\text{pr}_1} & \mathcal{M}_n \end{array}$$

where  $j \neq 1, 1728$  and where  $u$  and  $v$  are surjective étale, the projection  $\text{pr}_1$  is étale and, by faithfully flat descent,  $j$  is étale.

The section at infinity of the projective line  $\mathbb{P}_{\text{Spec}(\mathbb{Z}[1/n])}^1 \supset \mathbb{A}_{\text{Spec}(\mathbb{Z}[1/n])}^1$  over  $\text{Spec}(\mathbb{Z}[1/n])$  is a regular divisor, with generic point of characteristic 0, in a regular scheme. It then follows from a theorem of Abyankhar (see [5]) that, along this divisor  $j = \infty$ , the scheme  $\mathcal{M}_n$  is tamely ramified over  $\mathbb{P}^1$ , and that the normalization  $\mathcal{M}_n^*$  of  $\mathbb{P}^1$  in  $\mathcal{M}_n$  satisfies (4.1).  $\square$

From the same theorem, it follows that the  $\mathbb{Z}_\ell$ -sheaves l.c.c. on  $\mathcal{M}_n[1/\ell]$  are tamely ramified at infinity. Hence, from (4.1) and from the specialization theorems in  $\ell$ -adic cohomology (see [5]), it follows that  $R^i a_*(\mathcal{M}_n, \text{Sym}^k(R^1 f_* \mathbb{Z}_\ell))$ ,  $R^i a_!(\mathcal{M}_n, \text{Sym}^k(R^1 f_* \mathbb{Z}_\ell))$  and thus  $R^i \tilde{a}(\mathcal{M}_n, \text{Sym}^k(R^1 f_* \mathbb{Z}_\ell))$  are  $\mathbb{Z}_\ell$ -adic sheaves l.c.c. on  $\text{Spec}(\mathbb{Z}[1/n, 1/\ell])$ , whose formation is compatible with any base change.

**COROLLARY 4.2.** - The Galois module  ${}_n W_\ell$  is isomorphic to the fiber at the geometric point  $\overline{\mathbb{Q}}$  of  $\text{Spec}(\mathbb{Z}[1/n, 1/\ell])$  of the l.c.c.  $\mathbb{Q}_\ell$ -sheaf  $R^i \tilde{a}(\mathcal{M}_n, \text{Sym}^k(R^1 f_* \mathbb{Z}_\ell) \otimes \mathbb{Q}_\ell)$ . It is unramified outside of  $n$  and  $\ell$ .

Consider, over  $\mathcal{M}_n \otimes \mathbb{F}_p$ , the two commutative diagrams

$$\begin{array}{ccc} & (\mathbb{Z}/n)^2 & \\ \alpha \nearrow & & \nwarrow \alpha^{(p)} \\ E_n & \xrightarrow{\quad} & E_n^{(p)} \\ \downarrow & & \downarrow \\ E & \xrightarrow{F} & E^{(p)} \end{array} \quad \text{and} \quad \begin{array}{ccc} & (\mathbb{Z}/n)^2 & \\ p \cdot \alpha^{(p)} \nearrow & & \nwarrow \alpha \\ E_n^{(p)} & \xrightarrow{\quad} & E_n \\ \downarrow & & \downarrow \\ E^{(p)} & \xrightarrow{V} & E \end{array}$$

abbreviated as:

$$F : (E, \alpha) \longrightarrow (E^{(p)}, \alpha^{(p)}) \quad \text{and} \quad V : (E^{(p)}, p\alpha^{(p)}) \longrightarrow (E, \alpha) \quad ,$$

where  $F$  is the Frobenius morphism and  $V$ , its transpose, is the "Verschiebung". These diagrams define morphisms  $\Phi_1$  and  $\Phi_2$  from  $\mathcal{M}_n \otimes \mathbb{F}_p$  to  $\mathcal{M}_{n,p} \otimes \mathbb{F}_p$ . These morphisms are finite (as sections of  $q_1$  or  $q_2$ ) and define a morphism

$$\Phi = \Phi_1 \amalg \Phi_2 : \mathcal{M}_n \otimes \mathbb{F}_p \amalg \mathcal{M}_n \otimes \mathbb{F}_p \rightarrow \mathcal{M}_{n,p} \otimes \mathbb{F}_p.$$

Let  $\Phi^h$  be the restriction of  $\Phi$  to the open sets  $\mathcal{M}_n^h$  and  $\mathcal{M}_{n,p}^h$  of  $\mathcal{M}_n \otimes \mathbb{F}_p$  and  $\mathcal{M}_{n,p} \otimes \mathbb{F}_p$  which correspond to curves of nonzero Hasse invariant  $h$ .

**PROPOSITION 4.3.** — The morphism  $\Phi^h$  is an isomorphism.

Let  $\varphi : E_1 \rightarrow E_2$  be a  $p$ -isogeny between elliptic curves with invertible Hasse invariant on a scheme  $S$  of characteristic  $p$ . At each geometric point of  $S$ , either the kernel  $\text{Ker}(\varphi)$  of  $\varphi$  is étale over  $S$ , or its Cartier dual, isomorphic to  $\text{Ker}({}^t\varphi)$ , is étale over  $S$ . The property “ $\text{ker}(\varphi)$  is étale” is an open property, so that locally on  $S$  either  $\text{Ker}(\varphi)$  is purely infinitesimal or  $\text{Ker}({}^t\varphi)$  is infinitesimal. The only infinitesimal subgroup of order  $p$  of  $E_1$  or  $E_2$  being the kernel of Frobenius, in the first case,  $\varphi$  is isomorphic to  $F : E_1 \rightarrow E_1^{(p)}$  and in the second case,  ${}^t\varphi$  is isomorphic to  $F : E_2 \rightarrow E_2^{(p)}$  thus  $\varphi$  to  $V : E_2^{(p)} \rightarrow E_2$ .  $\square$

**PROPOSITION 4.4.**

- (i) The scheme  $\mathcal{M}_{n,p}$  is smooth over  $\text{Spec}(\mathbb{Z})$  outside the points of characteristic  $p$  where  $h = 0$ .
- (ii) The morphisms  $q_1$  and  $q_2$  induce finite and flat morphisms  $q'_1$  and  $q'_2$  from the normalization  $\mathcal{M}'_{n,p}$  of  $\mathcal{M}_{n,p}$  to  $\mathcal{M}_n$ .
- (iii) The morphism  $\Phi$  factors through a surjective morphism

$$\Phi' : \mathcal{M}_n \otimes \mathbb{F}_p \coprod \mathcal{M}_n \otimes \mathbb{F}_p \rightarrow \mathcal{M}'_{n,p} \otimes \mathbb{F}_p.$$

The automorphism  $\sigma$  of (3.15) exchanges  $\varphi$  and  ${}^t\varphi$ , so that it suffices to prove (i) at the points of characteristic  $p$  of  $\mathcal{M}_{n,p}$  where the kernel of  $\varphi$  is infinitesimal: there is no obstruction to lifting an elliptic curve infinitesimally and to lifting the infinitesimal part of the kernel of multiplication by  $p$ .

Where  $p = h = 0$ , the fiber of the finite morphism  $(3.15)q_i : \mathcal{M}_{n,p} \rightarrow \mathcal{M}_n$  is reduced to a point, so that the smooth locus of  $\mathcal{M}_{n,p}$  is dense in  $\mathcal{M}_{n,p}$  and  $\mathcal{M}'_{n,p}$  is everywhere of dimension 2. The scheme  $\mathcal{M}_n$  being regular, by EGA 0<sub>IV</sub> 16.5.1 and 17.3.5 (ii), the morphism  $q_i : \mathcal{M}'_{n,p} \rightarrow \mathcal{M}_n$  is flat. Finally, (iii) results from the fact that  $\Phi$  is finite and  $\mathcal{M}_n \otimes \mathbb{F}_p$  is a normal curve.  $\square$

The Hecke endomorphism  $T_p$  of  ${}_nW_\ell$ , as is explained in (3.18), is the  $\mathbb{Q}_\ell$ -tensor of the fiber at the geometric point  $\overline{\mathbb{Q}}$  of  $\text{Spec}(\mathbb{Z}[1/n, 1/\ell])$  of the endomorphism (again denoted  $T_p$ ) of

$$R^1\tilde{a}\left(\mathcal{M}_n, \text{Sym}^k(R^1f_{n*}(\mathbb{Z}_\ell))\right)$$

defined by the “correspondence”.

$$(4.5) \quad \begin{array}{ccccc} & q_1'^* \mathcal{E} & \xrightarrow{\varphi} & q_2'^* \mathcal{E} & \\ & \swarrow u & & \searrow v & \\ \mathcal{E} & & \mathcal{M}'_{n,p} & & \mathcal{E} \\ & \searrow f_n & & \swarrow f_n & \\ & \mathcal{M}_n & & \mathcal{M}_n & \end{array}$$

$$T_p = q_1'^* \varphi^* q_2'^* \quad (\text{cf. 3.18}).$$

The endomorphisms  $R_p$  and  $I_p$  are interpreted in a similar way.

**LEMMA 4.6.** - Let  $S$  be a noetherian scheme and let  $X, Y, Z_1, Z_2$  be four  $S$ -schemes, separated and of finite type; let  $F$  be a  $\mathbb{Z}_\ell$ -sheaf on  $X$ , and  $\underline{G}$  a  $\mathbb{Z}_\ell$ -sheaf on  $Y$ ; and denote by  $a$  each of the structural maps of  $X, Y, Z_1$  or  $Z_2$  into  $S$ .

Suppose given a commutative diagram of  $S$ -schemes, and morphisms of sheaves:

$$\begin{array}{ccc} y_1^* \underline{G} & \xrightarrow{z_1} & x_1^* \underline{F} \\ y_2^* \underline{G} & \xrightarrow{z_2} & x_2^* \underline{F} \end{array}$$
  

$$\begin{array}{ccccc} & & Z_1 & \xrightarrow{f} & Z_2 \\ & x_1 \swarrow & & \searrow \swarrow & \\ X & & & & Y \\ & \nwarrow x_2 & & \nwarrow y_1 & \end{array}$$

Assume that  $f^* z_2 = z_1$ , that  $y_1$  and  $y_2$  are proper, that  $x_1$  and  $x_2$  are finite and flat, and that for every geometric point  $s$  of  $Z_2$  the multiplicity of  $s$  in its fiber  $x_2^{-1}(x_2(s))$  is equal to the sum of the multiplicities in the fiber (for  $x_1$ ) of the geometric points of  $Z_1$  lying over  $s$  via  $f$ .

Then the diagram

$$\begin{array}{ccccccc} R^i \tilde{a}(Y, \underline{G}) & \xrightarrow{y_1^*} & R^i \tilde{a}(Z_1, \underline{G}) & \xrightarrow{z_1} & R^i \tilde{a}(Z_1, \underline{F}) & \xrightarrow{x_{1*}} & R^i \tilde{a}(X, \underline{F}) \\ \parallel & & \uparrow & & \uparrow & & \parallel \\ R^i \tilde{a}(Y, \underline{G}) & \xrightarrow{y_2^*} & R^i \tilde{a}(Z_2, \underline{G}) & \xrightarrow{z_2} & R^i \tilde{a}(Z_2, \underline{F}) & \xrightarrow{x_{2*}} & R^i \tilde{a}(X, \underline{F}) \end{array}$$

is commutative.

This lemma results from analogous lemmas for  $R^1 a_!$  and  $R^1 a_*$ . The commutativity of the first squares is trivial. The last square rewrites as

$$\begin{array}{ccccc} R^i \tilde{a}(Z_1, \underline{F}) & \xleftarrow{\sim} & R^i \tilde{a}(X, x_{1*} x_1^* \underline{F}) & \xrightarrow{\text{Tr}} & R^i \tilde{a}(X, \underline{F}) \\ \uparrow & & \uparrow & & \parallel \\ R^i \tilde{a}(Z_2, \underline{F}) & \xleftarrow{\sim} & R^i \tilde{a}(X, x_{2*} x_2^* \underline{F}) & \xrightarrow{\text{Tr}} & R^i \tilde{a}(X, \underline{F}) \end{array}$$

and one returns to the definition of the trace to verify that the square

$$\begin{array}{ccc} x_{1*} x_1^* \underline{F} & \xrightarrow{\text{Tr}} & \underline{F} \\ \uparrow & & \parallel \\ x_{2*} x_2^* \underline{F} & \xrightarrow{\text{Tr}} & \underline{F} \end{array}$$

commutes.

(4.7) We denote by  $T_p/\mathbb{F}_p$  the endomorphism induced by  $T_p$  on the restriction to  $\text{Spec}(\mathbb{F}_p)$  of the l.c.c.  $\mathbb{Z}_\ell$ -sheaf  $R^1 \tilde{a}(\mathcal{M}_n, \text{Sym}^k(R^1 f_{n*} \mathbb{Z}_\ell))$ . We have

$$R^1 \tilde{a}(\mathcal{M}_n, \text{Sym}^k(R^1 f_{n*} \mathbb{Z}_\ell))|_{\text{Spec}(\mathbb{F}_p)} \simeq R^1 \tilde{a}(\mathcal{M}_n \otimes \mathbb{F}_p, \text{Sym}^k(R^1 f_{n*} \mathbb{Z}_\ell));$$

The formation of the trace morphism for a finite and flat morphism is compatible with base change, so that one may construct, on the model (3.18), from the fiber over  $\mathbb{F}_p$  of the “correspondence” (4.5). Lemma (4.6), applied to the commutative diagram

$$\begin{array}{ccccc}
 & \mathcal{M}_n \otimes \mathbb{F}_p \amalg \mathcal{M}_n \otimes \mathbb{F}_p & \xrightarrow{\Phi'} & \mathcal{M}'_{n,p} \otimes \mathbb{F}_p & \\
 & \swarrow & & \searrow & \\
 \mathcal{M}_n \otimes \mathbb{F}_p & & \xleftarrow{q'_1} & & \xrightarrow{q'_2} \mathcal{M}_n \otimes \mathbb{F}_p
 \end{array}$$

then provides a decomposition of  $T_p/\mathbb{F}_p$  as the sum of the endomorphisms defined by the following two correspondences:

$$\begin{array}{ccccc}
 & (\mathcal{E}, \alpha) & \xrightarrow{\quad} & (\mathcal{E}^{(p)}, \alpha^{(p)}) = F^*(\mathcal{E}, \alpha) & \\
 & \swarrow & & \searrow & \\
 \text{(a)} & (\mathcal{E}, \alpha) & & \mathcal{M}_n \otimes \mathbb{F}_p & \xrightarrow{F} \mathcal{M}_n \otimes \mathbb{F}_p \\
 & \searrow & & \swarrow & \\
 & \mathcal{M}_n \otimes \mathbb{F}_p & \xrightarrow{\quad} & \mathcal{M}_n \otimes \mathbb{F}_p & \xrightarrow{\quad} (\mathcal{E}, \alpha)
 \end{array}$$

where  $F$  is the absolute Frobenius. One recognizes in this correspondence the geometric Frobenius.

$$\begin{array}{ccccc}
 & x^*(\mathcal{E}, \alpha) = (\mathcal{E}^{(p)}, p\alpha^{(p)}) & \xrightarrow{V} & (\mathcal{E}, \alpha) & \\
 & \swarrow & & \searrow & \\
 \text{(b)} & (\mathcal{E}, \alpha) & & \mathcal{M}_n \otimes \mathbb{F}_p & \xrightarrow{f_n} \mathcal{M}_n \otimes \mathbb{F}_p \\
 & \searrow & & \swarrow & \\
 & \mathcal{M}_n \otimes \mathbb{F}_p & \xrightarrow{x} & \mathcal{M}_n \otimes \mathbb{F}_p & \xrightarrow{\quad} (\mathcal{E}, \alpha)
 \end{array}$$

The map  $x$  is the composite map  $I_p^{-1} \circ F$ :

$$\begin{array}{ccccc}
 (\mathcal{E}, \alpha) & \longleftarrow & (\mathcal{E}, p\alpha) & \longleftarrow & (\mathcal{E}^{(p)}, p\alpha^{(p)}) \\
 \downarrow & & \downarrow & & \downarrow \\
 \mathcal{M}_n \otimes \mathbb{F}_p & \xleftarrow{I_p^{-1}} & \mathcal{M}_n \otimes \mathbb{F}_p & \xleftarrow{F} & \mathcal{M}_n \otimes \mathbb{F}_p
 \end{array}$$

The corresponding endomorphism is then the composition of

$$\begin{aligned}
 V : R^1 \tilde{a} \left( \mathcal{M}_n \otimes \mathbb{F}_p, \text{Sym}^k (R^1 f_{n*} \underline{\mathbb{Z}}_\ell) \right) &\xrightarrow{V^*} R^1 \tilde{a} \left( \mathcal{M}_n \otimes \mathbb{F}_p, \text{Sym}^k (R^1 f_{n*}^{(p)} \underline{\mathbb{Z}}_\ell) \right) \\
 &\xrightarrow{\text{Tr}_F} R^1 \tilde{a} \left( \mathcal{M}_n \otimes \mathbb{F}_p, \text{Sym}^k (R^1 f_{n*} \underline{\mathbb{Z}}_\ell) \right)
 \end{aligned}$$

and of

$$I_p^* = \text{Tr}_{I_p^{-1}} : \text{endomorphism of } R^1 \tilde{a} \left( \mathcal{M}_n \otimes \mathbb{F}_p, \text{Sym}^k (R^1 f_{n*} \underline{\mathbb{Z}}_\ell) \right).$$

**PROPOSITION 4.8.** — We have  $T_p/\mathbb{F}_p = F + I_p^* V$ , and

(i)  $F$  is identified with the inverse of the Frobenius element (“arithmetic”)  $\varphi_p$  of the Galois

- group  $\text{Gal}(\bar{\mathbb{F}}_p/\mathbb{F}_p)$  acting on  $\tilde{H}^1(\mathcal{M}_n \otimes \mathbb{F}_p, \text{Sym}^k(R^1 f_{n*} \underline{\mathbb{Z}}_\ell))$ ;
- (ii)  $F$  and  $V$  are transpose with respect to the scalar product (3.20);
- (iii)  $FV = VF = p^{k+1}$ .

For the relation (i) between the geometric and arithmetic Frobenius, see the exposition of C. Houzel (SGA 5, XV). The composite  $VF$  is the composite of the homomorphisms deduced from the following morphisms:

$$\begin{array}{ccccccc}
 \mathcal{E} & \longleftarrow & \mathcal{E}^{(p)} & \xrightarrow{F_{\mathcal{E}}} & \mathcal{E} & \xrightarrow{V_{\mathcal{E}}} & \mathcal{E}^{(p)} \longrightarrow \mathcal{E} \\
 \downarrow & & & \searrow & \downarrow & \swarrow & \\
 \mathcal{M}_n & \xleftarrow{F} & & & \mathcal{M}_n & \xrightarrow{F} & \mathcal{M}_n
 \end{array}$$

$$VF = \text{Tr}_F \circ F_{\mathcal{E}}^* \circ V_{\mathcal{E}}^* \circ F^*.$$

The morphism  $F_{\mathcal{E}}^* V_{\mathcal{E}}^* = (F_{\mathcal{E}} V_{\mathcal{E}})^* = (p \cdot 1_E)^*$  acts by multiplication by  $p^k$  on  $\text{Sym}^k(R^1 f_{*} \underline{\mathbb{Z}}_\ell)$ , so that  $VF = p^k \cdot \text{Tr}_F \circ F^* = p^k \cdot p = p^{k+1}$ , since  $F : \mathcal{M}_n \rightarrow \mathcal{M}_n$  is of degree  $p$ .

By transport of structure,  $\varphi_p$  respects the scalar product (3.20) taking values in  $\mathbb{Q}_\ell(-k-1)$ , a group on which  $\varphi_p$  acts by multiplication by  $p^{-k-1}$ . Hence one has

$$(Fx, y) = p^{k+1}(\varphi_p Fx, \varphi_p y) = (x, p^{k+1} F^{-1} y) = (x, Vy). \quad \square$$

The following theorem, synonymous with (4.8), goes back to Eichler.

**THEOREM 4.9 (Congruence Formula).** — Let  $K_{n,\ell}$  be the largest subextension of  $\mathbb{Q}$  unramified outside  $n$  and  $\ell$ , and let  $\varphi_p$  be a Frobenius element relative to  $p$  in  $\text{Gal}(K_{n,\ell}/\mathbb{Q})$ . Let  $F$  be the endomorphism  $\varphi_p^{-1}$  of  $W_\ell$  and  $V$  its transpose with respect to the scalar product (3.20). Then,

$$T_p = F + I_p^* V, \quad FV = p^{k+1}$$

and

$$1 - T_p X + p R_p X^2 = (1 - FX)(1 - I_p^* V X). \quad \square$$

## Chapter 5

# Weil implies Ramanujan

### No. 5 – Weil implies Ramanujan.

If  $p$  is a prime number and  $X$  is a scheme over  $\mathbb{F}_p$ , we denote by  $\overline{\mathbb{F}}_p$  an algebraic closure of  $\mathbb{F}_p$ , by  $F : X \rightarrow X$  the (geometric) Frobenius endomorphism, and we set  $\overline{X} = X \otimes \overline{\mathbb{F}}_p$ . Throughout,  $\ell$  will always denote a prime number distinct from  $p$ .

By "Weil conjectures" we mean the following statement:

Let  $X$  be a projective and smooth scheme over  $\mathbb{F}_p$  and let  $\ell$  be a prime different from  $p$ . Then the eigenvalues of the endomorphism  $F^*$  on  $H^i(X, \mathbb{Q}_\ell)$  are algebraic integers, all of whose complex conjugates have absolute value  $p^{i/2}$ .

With the hypotheses and notations of (4.9) (recall that  $(p, n) = 1$ ), we have:

**THEOREM 5.1.** - If the Weil conjectures are true, then the eigenvalues of the endomorphism  $F$  of  $W_\ell$  are algebraic integers (all of whose complex conjugates have absolute value  $p^{(k+1)/2}$ ).

Admit the Weil conjectures.

**LEMMA 5.2.** (modulo Weil).- Let  $X$  be a smooth scheme over  $\mathbb{F}_p$  which can be represented as an open subset of a projective smooth scheme  $X^*$ . Then the eigenvalues of the endomorphism  $F^*$  on  $\tilde{H}^i(X, \mathbb{Q}_\ell)$  are algebraic integers of absolute value  $p^{i/2}$ .

The natural map from  $H_c^i(\overline{X}, \mathbb{Q}_\ell)$  to  $H^i(\overline{X}, \mathbb{Q}_\ell)$  factors through  $H^i(\overline{X}^*, \mathbb{Q}_\ell)$ :

$$H_c^i(\overline{X}, \mathbb{Q}_\ell) \rightarrow H^i(\overline{X}^*, \mathbb{Q}_\ell) \rightarrow H^i(X, \mathbb{Q}_\ell)$$

so that as a  $\text{Gal}(\overline{\mathbb{F}}_p/\mathbb{F}_p)$ -module,  $\tilde{H}^i(\overline{X}, \mathbb{Q}_\ell)$  is a subquotient of  $H^i(\overline{X}^*, \mathbb{Q}_\ell)$ .  $\square$

**LEMMA 5.3** (modulo Weil).- Let  $S$  be a smooth scheme over  $\mathbb{F}_p$  and let  $f : A \rightarrow S$  be an abelian scheme over  $S$ . Suppose that  $A$  can be represented as an open subset of a projective smooth scheme  $A'$  over  $\mathbb{F}_p$ . Then the geometric Frobenius endomorphism  $F^*$  of  $\tilde{H}^i(\overline{S}, R^j f_* \mathbb{Q}_\ell)$  has eigenvalues which are algebraic integers of absolute value  $p^{(i+j)/2}$ .

Let  $m > 1$  be an integer, and consider the Leray spectral sequence

$$\begin{aligned} E : E_2^{ij} &= H^i(\bar{S}, R^j f_* \mathbb{Q}_\ell) \Rightarrow H^{i+j}(\bar{A}, \mathbb{Q}_\ell) \\ {}_c E : {}_c E_2^{ij} &= H_c^i(\bar{S}, R^j f_* \mathbb{Q}_\ell) \Rightarrow H^{i+j}(\bar{A}, \mathbb{Q}_\ell). \end{aligned}$$

The endomorphism of multiplication by  $m : \psi_m = m1_A$ , defines endomorphisms of  $E$  and  ${}_c E$  which are inserted into a commutative diagram:

$$\begin{array}{ccc} {}_c E & \longrightarrow & E \\ \downarrow \psi_m^* & & \downarrow \psi_m^* \\ {}_c E & \longrightarrow & E \end{array}$$

On  $R^j f_* \mathbb{Q}_\ell$ ,  $\psi_m^*$  acts by multiplication by  $m^j$ , such that on the terms  ${}_c E_r^{ij}$  and  $E_r^{ij}$  of  ${}_c E$  and  $E$ ,  $\psi_m^*$  is the multiplication by  $m^j$ . The maps  $d_r$  ( $r \geq 2$ ) commutes with  $\psi_m^*$ , and send  $E_r^{ij}$  (resp.  ${}_c E_r^{ij}$ ) into  $E_r^{i'j'}$  (resp.  ${}_c E_r^{i'j'}$ ) with  $j \neq j'$ . They are therefore 0, and  $E_2^{ij}$  (resp.  ${}_c E_2^{ij}$ ) is identified with the subspace of  $H^{i+j}(\bar{A}, \mathbb{Q}_\ell)$  (resp. of  $H_c^{i+j}(\bar{A}, \mathbb{Q}_\ell)$ ) where  $\psi_m^* = m^j$ . Therefore,  $\tilde{H}^i(\bar{S}, R^j f_* \mathbb{Q}_\ell)$  is identified with the galois submodule of  $\tilde{H}^{i+j}(\bar{A}, \mathbb{Q}_\ell)$  where  $\psi_m^* = m^j$  and we apply (5.2). The trick used here is due to Lieberman.  $\square$

Let  $f_n : \mathcal{E} \rightarrow \mathcal{M}_n \otimes \mathbb{F}_p$  be the universal elliptic curve on  $\mathcal{M}_n \otimes \mathbb{F}_p$  and let  $f_{n,k} : \mathcal{E}_k \rightarrow \mathcal{M}_n \otimes \mathbb{F}_p$  be its iterated  $k$ -fold fiber product with itself. The Kunneth's formula shows that the  $\mathbb{Q}_\ell$ -sheaf  $R^k f_{n,k*} \mathbb{Q}_\ell$  admits as direct factor the  $k$ -th tensor power of  $R^1 f_{n*} \mathbb{Q}_\ell$ ; this in turn contains as direct factor the  $\mathbb{Q}_\ell$ -sheaf  $\text{Sym}^k(R^1 f_{n*} \mathbb{Q}_\ell)$ . Theorem 5.1 is thus a result of (5.3) and of

**LEMMA 5.4.** - The scheme  $\mathcal{E}^{(k)}$  is an open subset of a scheme  $\mathcal{E}^*$  which is projective and smooth over  $\mathbb{F}_p$ .

Let  $\mathcal{E}^*$  be the minimal Néron model of  $\mathcal{E}$  over  $\mathcal{M}_n^* \otimes \mathbb{F}_p$  (4.1). The scheme  $\mathcal{E}^*$  is projective and smooth over  $\mathbb{F}_p$ . Since  $n \geq 3$  and since the  $n$ -torsion points of  $\mathcal{E}$  form a trivial covering of  $\mathcal{M}_n \otimes \mathbb{F}_p$ , this Néron model is “semi-stable” (case  $a$  or  $b_m$  in Néron's classification). In particular, the projection  $f : \mathcal{E}^* \rightarrow \mathcal{M}_n^*$  has only finitely many non-smooth points, and at these points  $f_n$  is non-degenerate (exhibiting an ordinary quadratic singularity).

Let  $\mathcal{E}_k^{**}$  be the  $k$ -th iterated fiber product of  $\mathcal{E}^*$  over  $\mathcal{M}_n^*$ . To prove (5.4), it suffices to resolve the singularity of  $\mathcal{E}_k^{**}$  without touching the open subset  $\mathcal{E}_k$ . Let's prove first:

**LEMMA 5.5.** - Let  $V$  be the subvariety of the affine space over a field  $k$  (with coordinates  $X_0, Y_0, \dots, X_r, Y_r, T_1 \dots T_s$ ) defined by the equations

$$X_0 Y_0 = X_1 Y_1 = \dots = X_r Y_r.$$

Let  $m$  be the ideal of  $\mathcal{O}_V$  generated by the monomials obtained from the monomials deduced from  $\prod_{i=0}^r X_i^i$  by any permutation of the coordinates that respects the set of pairs  $\{X_i, Y_i\}$  (for  $0 \leq i \leq r$ ). Then,  $m = \mathcal{O}_V$  outside the singular locus of  $V$ , and the variety  $\tilde{V}$  obtained from  $V$  by blowing up the ideal  $m$  is smooth over  $k$ .

The singular locus is the locus where, for some  $i \neq j$ , the four coordinates  $X_i, Y_i, X_j, Y_j$  vanish simultaneously. The affine open subset of  $\tilde{V}$  defined by the element  $\prod_1^r X_i^i$  of the ideal  $m$  is the spectrum of the regular ring

$$k \left[ Y_0/X_1, X_0/X_1, X_1/X_2, \dots, X_{r-1}/X_r, X_r, T_1, \dots, T_s \right].$$

(To verify, note that  $X_i/X_{i+1} = Y_{i+1}/Y_i$ ), and Lemma 5.5 follows.  $\square$

One now shows that, locally for the étale topology, the singularities of  $\mathcal{E}_k^{**}$  are isomorphic to those of  $V$  (with  $r = k - 1$ ), and that this permits one to define on  $\mathcal{E}_k^{**}$  an ideal  $m$  analogous to the ideal  $m$  in Lemma 5.5. Blowing up this ideal yields  $\mathcal{E}_k^*$ .  $\square$

An approximation of the following theorem has been proved by Ihara [2]:

**THEOREM 5.6.** - (Weil Implies Ramanujan.) The Weil conjectures imply the Ramanujan conjecture.

Note first that (5.1) remains true for  $n = 1$ , because  ${}^k_1W_\ell$  is the galois submodule of  ${}_mW_\ell^k$  that is invariant under  $\mathrm{GL}_2(\mathbb{Z}/m\mathbb{Z})$ . On  ${}_1W_\ell^k$ ,  $I_p^*$  induces the identity, and (4.8) reduces to

$$1 - T_p X + p^{k+1} X^2 = (1 - FX)(1 - VX).$$

The endomorphisms  $F$  and  $V$  are transposed with respect to one another, so that

$$\det(1 - FX; {}^k_1W_\ell) = \det(1 - VX; {}^k_1W_\ell).$$

The action of  $T_p$  on  ${}^k_1W_\ell$  is induced by its action on  ${}^k_1W$  and is compatible with the decomposition of  ${}^k_1W \otimes \mathbb{C}$  into the direct sum of the space  $S_{k+2}$  of parabolic modular forms of weight  $k + 2$  for  $SL_2(\mathbb{Z})$  and its complex conjugate. Since  $T_p$  is a hermitian operator (for the Petersson scalar product) and (3.19), one then deduces that

$$\det(1 - T_p X + p^{k+1} X^2; {}^k_1W_\ell) = \det(1 - T_p X + p^{k+1} X^2; S_{k+2})^2,$$

and

$$\det(1 - T_p X + p^{k+1} X^2; S_{k+2})^2 = \det(1 - FX; {}^k_1W_\ell)^2$$

i.e.

$$(5.7) \quad \det(1 - T_p X + p^{k+1} X^2; S_{k+2}) = \det(1 - FX; {}^k_1W_\ell).$$

Returning to the notations of chapter 1 and taking  $k = 10$ , by Hecke's theory and (3.19), (5.7) is rewritten as

$$H_p(X) = \det(1 - FX; {}^{10}_1W_\ell)$$

and one applies (5.1).  $\square$

One similarly verifies that the Weil conjectures imply the generalization by Petersson of the Ramanujan conjecture.

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